Problem 5

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Problem 1 *True/False: Give an explanation or counterexample. Assume a and L are finite numbers.*

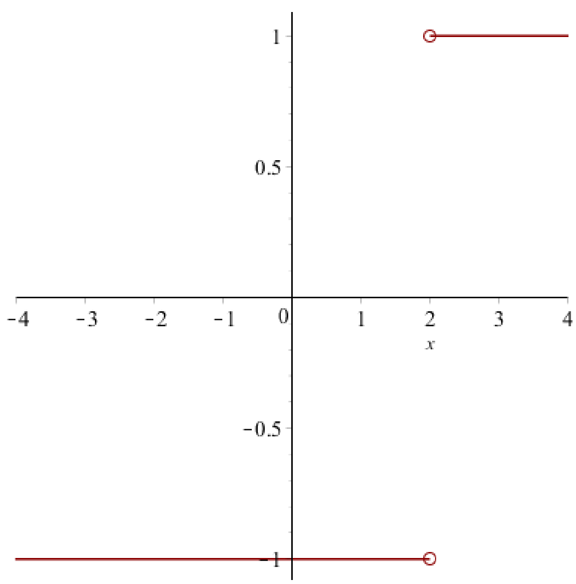
(a) *If* $\lim_{x\to a} f(x) = L$, *then* $f(a) = L$.

Explanation. False. In the graph below $\lim_{x\to 1} f(x) = 2$, but $f(1)$ does not exist.

alt=graph of function with f(x)=x+1, for x not equal to 1. it has an open circle at the point (1, 2)

(b) *If* $\lim_{x\to a^-} f(x) = L$, *then* $\lim_{x\to a^+} f(x) = L$.

Explanation. False. In the graph below $\lim_{x\to 2^-} f(x) = -1$ but $\lim_{x\to 2^+} f(x) = 1$.



(c) *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = L$, *then* $f(a) = g(a)$.

Explanation. False. If we let

alt=graph of a line and parabola, with open circle at their intersection. points are filled in above the intersection for the line and below the intersection for the parabola.

we see that $\lim_{x\to 1} f(x) = \lim_{x\to 1} g(x) = 2$, but $f(1) \neq g(1)$.

(d) $\lim_{x\to a} \frac{f(x)}{g(x)}$ *does not exist if* $g(a) = 0$.

Explanation. False. If $f(x) = 5x^2$ and $g(x) = x^2$, then $g(0) = 0$ but

$$\lim_{x\to 0} \frac{f(x)}{g(x)} = \lim_{x\to 0} \frac{5x^2}{x^2} = \lim_{x\to 0} 5 = 5.$$